IoT Internet of Things

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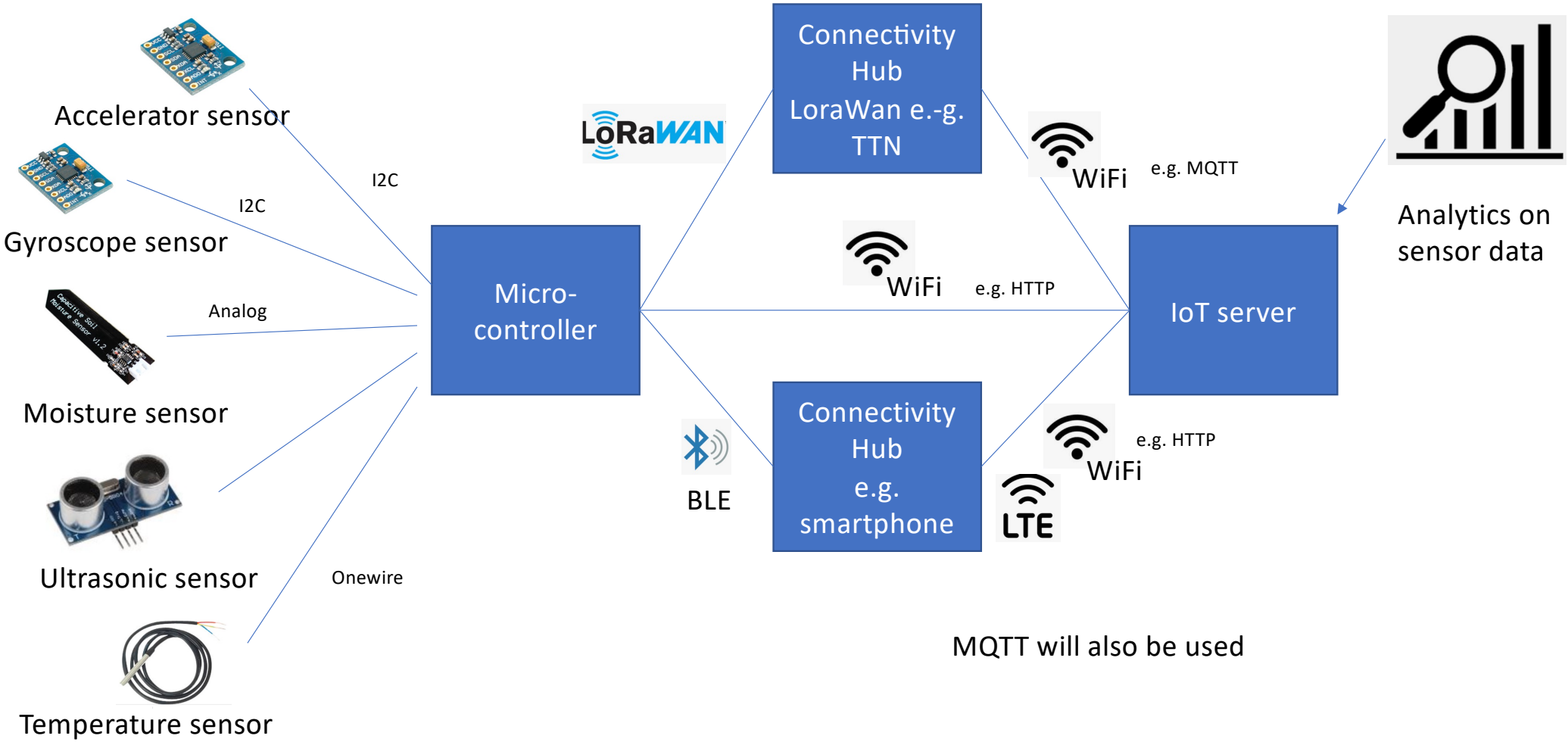
Block diagram IoT Lab – the architecture overview diagram we discussed so far



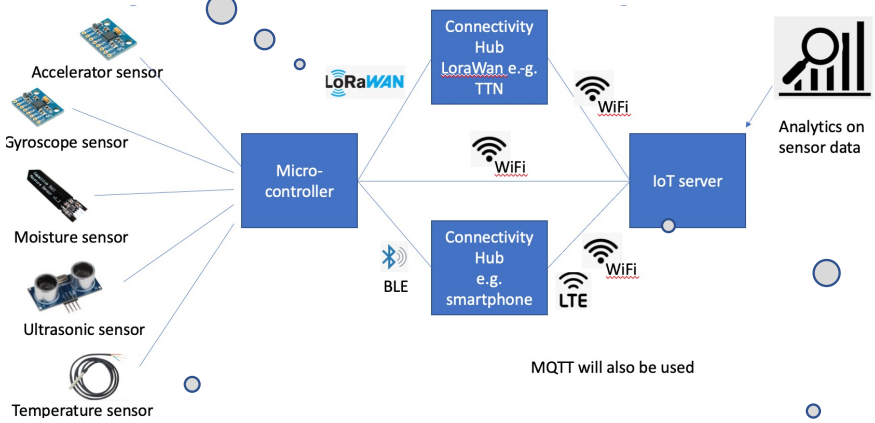
More topics to consider for an ‘End to End’ IoT Solution environment

IOT user

Data flow
Data values

Non functional requirements

Register and Monitor End Nodes



Analytics
AI
Machine learning

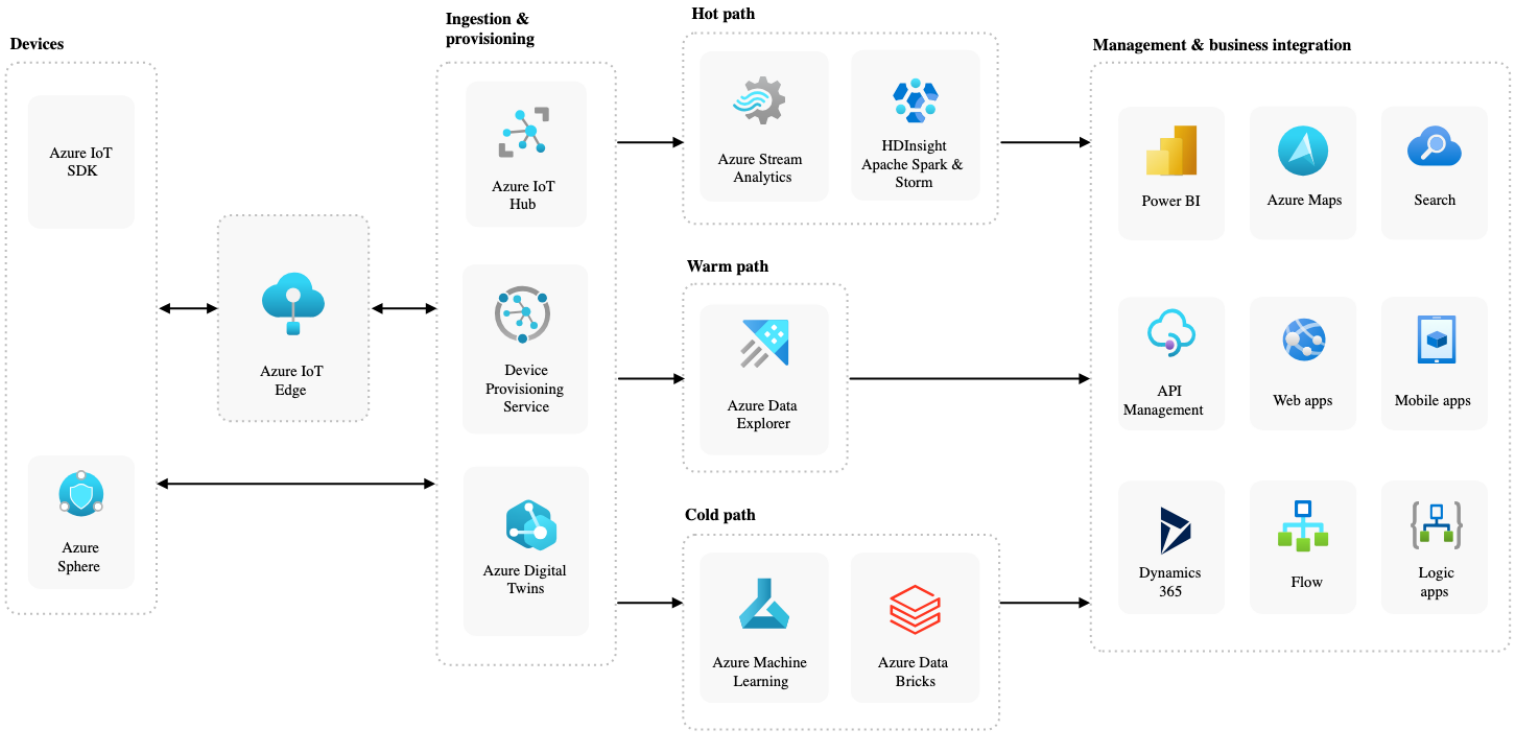
Storing the data for analytics ‘data lake’

Software updates End Nodes

IoT Security and IoT Governance

Different types of data Timeseries data, Geodata, in motion data, telematic data

Let’s have a look to the Azure IoT Reference Architecture



Microsoft Azure

The ‘device’ side

- Azure IoT Device
  - Azure IoT supports a large range of devices, from microcontrollers running Azure RTOS and Azure Sphere to developer boards like MX Chip and Raspberry Pi. Azure IoT also supports smart server gateways capable of running custom code. Devices might perform some local processing through a service such as Azure IoT Edge, or just connect directly to Azure so that they can send data to and receive data from the IoT solution.

- Azure IoT Hub
  - Azure IoT Hub is a managed service hosted in the cloud that acts as a central message hub for communication between an IoT application and its attached devices. You can connect millions of devices and their backend solutions reliably and securely. Almost any device can be connected to an IoT hub.

- Azure IoT Hub Device Provisioning Service
  - Microsoft Azure provides a rich set of integrated public cloud services for all your IoT solution needs. The IoT Hub Device Provisioning Service (DPS) is a helper service for IoT Hub that enables zero-touch, just-in-time provisioning to the right IoT hub without requiring human intervention. DPS enables the provisioning of millions of devices in a secure and scalable manner.

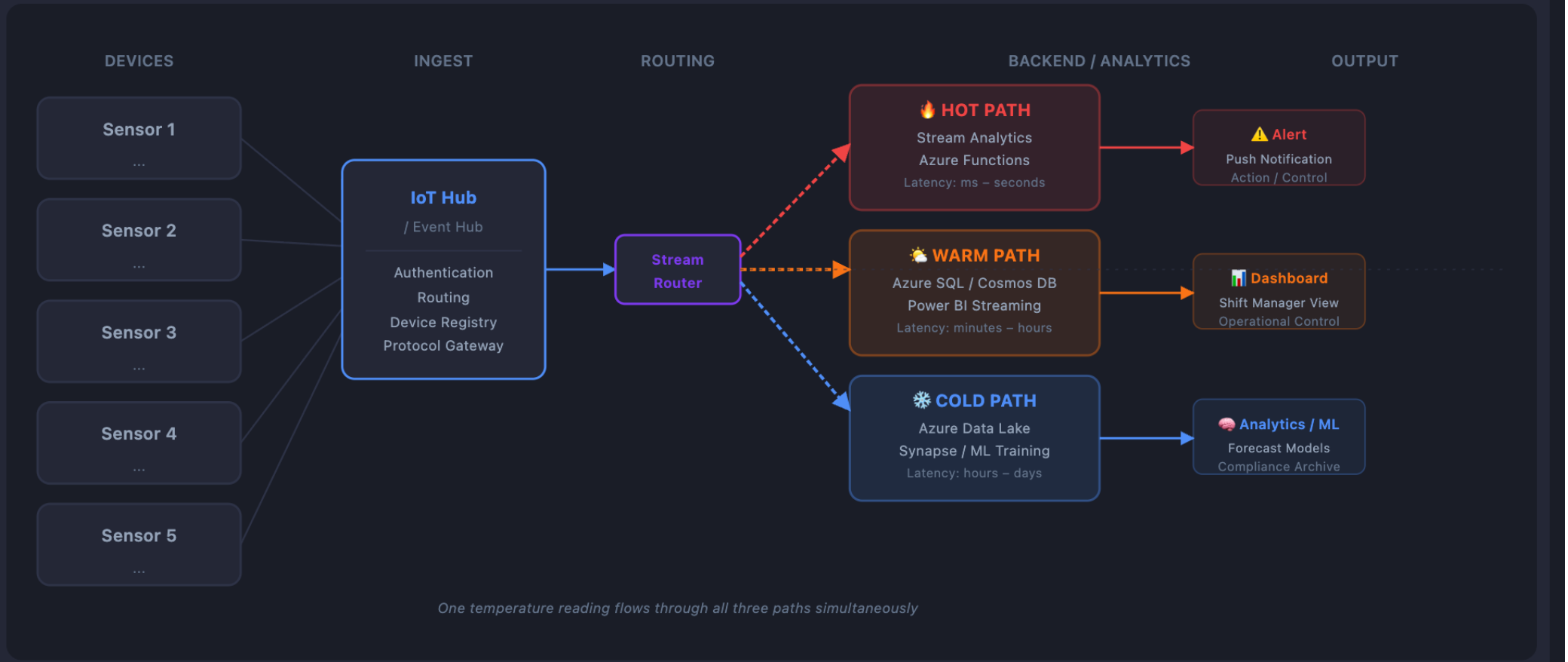
The ‘application’ side

- Azure IoT hot path
  - The **hot path** analyzes data in near-real-time as it arrives. Hot path telemetry must be processed with very low latency. The hot path typically uses a stream processing engine. Consider using services such as Azure Stream Analytics or Azure HDInsight. The output might trigger an alert, or be written to a structured format that can be queried using analytical tools.

- Azure IoT warm path
  - The **warm path** analyzes data that can accommodate longer delays for more detailed processing. Consider Azure Data Explorer for storing and analyzing large volumes of data.

- Azure IoT cold path
  - The **cold path** performs batch processing at longer intervals, like hourly or daily. The cold path typically operates over large volumes of data, which can be stored in Azure Data Lake Storage. Results don't need to be as timely as in the hot or warm paths. Consider using Azure Machine Learning or Azure Databricks to analyze cold data.

OVERALL ARCHITECTURE – DATA FLOW



Use cases - Hot Path Data in Motion – Real-Time

ms – SecondsStream AnalyticsAzure Functions
Raw data directly from the device, without buffering. The Hot Path decides – it triggers alerts and automated actions before data is ever stored.

REAL-WORLD USE CASES

- **CNC Machine Emergency Stop** (Manufacturing) – Vibration sensor exceeds 15 mm/s → spindle shuts down in <50 ms to prevent tool breakage
- **Pipeline Pressure Drop** (Oil & Gas) – Pressure falls below threshold → automatic valve closure + control room alert within 2 seconds
- **Cardiac Arrhythmia Detection** (Healthcare) – ECG monitor detects irregular pattern → nurse call system + defibrillator standby triggered instantly
- **Grid Voltage Spike** (Energy) – Substation sensor reads overvoltage → automatic circuit breaker trips within milliseconds
- **Fire Detector Activation** (Smart Building) – Smoke sensor threshold crossed → evacuation alarm + fire brigade dispatch in <10 seconds

Use cases - Warm Path Near Real-Time Analytics

Minutes – HoursAzure SQLPower BI

Aggregated or lightly filtered data, quickly queryable. The Warm Path informs people – operators and managers get a current operational picture.

REAL-WORLD USE CASES

- **Wind Farm Output Dashboard** (Energy) – Per-turbine power generation and efficiency, refreshed every 2 minutes for plant operators
- **Airport Gate Status Board** (Aviation) – Passenger flow, gate occupancy, boarding status across all terminals, updated every 3 minutes
- **Retail Shelf Replenishment View** (Retail) – Weight sensors on shelves aggregate stock levels per category, alerts buyer when <20% full
- **City Traffic Flow Heatmap** (Smart City) – Loop detectors and cameras feed a congestion map for the traffic control center, 5-min refresh
- **Data Center Power & Thermal View** (IT) – Rack-level temperature and power draw aggregated per row, flagging hotspots for operations team

Use cases - Cold Path Data at Rest – Historical

Hours – DaysData LakeSynapse / ML

Complete raw data, archived long-term. The Cold Path learns from the past – the foundation for forecasts, optimization, and compliance evidence.

REAL-WORLD USE CASES

- **Predictive Bearing Failure** (Manufacturing) – 6 months of motor vibration history trains ML model to predict bearing failure 2 weeks in advance
- **Grid Demand Forecasting** (Energy) – Historical load + weather + holiday data → 30-day electricity demand forecast for procurement
- **Telematics Risk Scoring** (Insurance) – Vehicle speed, braking, and cornering history → individual driver risk model for premium calculation
- **Crop Yield Optimization** (Agriculture) – Multi-season soil moisture, pH, and weather data → ML model recommends irrigation and fertilization schedules
- **CO_2 Emissions Report** (Compliance) – Full year of factory sensor data aggregated into annual regulatory emissions report (ISO 14064)

DECISION MATRIX – WHICH PATH AND WHEN?

CRITERION	🔥 HOT PATH	🌤️ WARM PATH	❄️ COLD PATH
Latency	**ms – Seconds**	**Minutes – Hours**	**Hours – Days**
Data Volume	Individual events, streams	Aggregates, time windows	Full archive, raw data
Goal	**Alert, Control**	**Inform, Monitor**	**Analyze, Learn**
Storage	None / short-lived (seconds)	DB, fast queryable	Data Lake, low cost
Azure Services	Stream Analytics, Functions	SQL DB, Cosmos DB, Power BI	Blob Storage, Data Lake, Synapse, ML
User	Machine / Automation	Shift Manager, Dispatcher	Analyst, ML Model, Auditor
Industry Example	**Pipeline valve closure <2 sec** *(Oil & Gas)*	**Wind farm output dashboard** *(Energy)*	**Predictive bearing failure ML** *(Manufacturing)*
Cost / Byte	High (compute-intensive)	Medium	Low (Blob/Parquet)

End to End point of view
Walk through – The device itself and the device registration

- **An IoT device** (source: Microsoft azure)
  - Process an unique identity that distinguishes it within the solution
  - Has properties, or state that the application can access
  - Sends events to the IoT platform for the application to act on
  - Receives commands from the application to act on

- **Device Metadata and device registration (these are sample attributes)**
  - Device Type
  - Type of Sensor
  - Settings e.g. which parameter the operator can set
  - Measurement e.g. telemetry the device sends
  - Location Info
  - Rules and Actions e.g. to enable operators to monitor the device
  - Hardware and software version
  - Value range
  - Activating / onboarding the device (e.g. OTAA over the air activation / ABP activation by personalization) – Initial key settings
  - …

➢ **An application component is required to cover this functionality**

Use cases cross industry and ‘path alignment’

CNC Emergency Stop *Manufacturing* HOT	Pipeline Valve Closure *Oil & Gas* HOT	Cardiac Arrhythmia Alert *Healthcare* HOT	Grid Overvoltage Breaker *Energy* HOT	Fire & Evacuation Alarm *Smart Building* HOT
Wind Farm Output Dashboard *Energy* WARM	Airport Gate Status Board *Aviation* WARM	Shelf Replenishment View *Retail* WARM	City Traffic Heatmap *Smart City* WARM	Data Center Thermal View *IT Operations* WARM
Predictive Bearing Failure *Manufacturing* COLD	Grid Demand Forecasting *Energy* COLD	Telematics Risk Scoring *Insurance* COLD	Crop Yield Optimization *Agriculture* COLD	CO_2 Emissions Report *Compliance* COLD

End to End point of view
Device monitoring and measurement monitoring

- **Is device monitoring required?**
  - **Keep alive ping**
  - **Watchdog functionality**
  - **Remote reset**
  - **Does the device operate differently from ‘normal operation’**
- **Is measurement monitoring required?**
  - **What to do if no data arrive**
  - **How to handle missing values**
  - **How to handle out of range values**
  - **…..**
- **Alerting**
  - **When something occurs which should be reported**

➢ **An application component is required to cover this functionality**

End to End point of view Walk through – Software updates and ‘OTAA over the air activation’

- **Device activation over the air (remember IoT Security – LoRaWAN example)**
- **Software updates over the air?**
  - **Software updates over the air**
    - **Memory available for new download**
    - **Fall back scenario**

➢ **An application component is required to cover this functionality**

End to End point of view
Walk through – IoT Security – IoT Governance

- **Authentication**
  - Device Authentication - This involves verifying the identity of IoT devices before allowing them to connect to a network or communicate with other devices.

- **Integrity**
  - Data integrity ensures that the information collected by IoT devices is accurate and has not been tampered.
  - Methods; checksums, hashes, and cryptographic signatures

- **Confidentiality**
  - Confidentiality ensures that data collected and transmitted by IoT devices is kept secure and not exposed to unauthorized individuals or entities.
  - Method: Using secure communication protocols (like HTTPS, MQTT with TLS, or CoAP over DTLS)

- **Robustness**
  - How fault tolerant the system will be. Redundency (e.g. backup power supplies)…..

End to End point of view
Walk through – IoT Security – IoT Governance

- **Availability**
  - Availability is often quantified as a percentage representing the time during which devices and services are operational. E.g. 99,1%
- **Reliability**
  - Reliable IoT systems consistently perform their tasks correctly, providing accurate data and responses under normal and varying operating conditions.
- **Interoperability**
  - Interoperability often relies on the establishment of common standards and protocols that enable devices from different manufacturers to communicate effectively.
  - Examples include MQTT, CoAP, and HTTP/REST for messaging,
  - as well as standards like Zigbee, Z-Wave, and Bluetooth for wireless communication and Thread / Matter
- **Accessability**

End to End point of view
Walk through – What to do with the data received

- **Data lake (big data)**
  - **Format of the data**
  - **Type of the data**

    - **Timeseries data**
      - **Characteristic: Sequence of values or events that are recorded in specific time intervals. Each data value is associated with a time stamp.**
      - **Time series data often show**
        - **Trends**
        - **Seasonality**
        - **Cycling Patterns**
        - **Stationarity**
        - **Autocorrelation**
        - **Noise**
        - **Irregular Values**

      - **Very often are timeseries data stored in timeseries databases**

End to End point of view
Walk through – What to do with the data received

- **Data lake (big data)**

  - **Geolocation data**
    - Geolocation data refers to information that identifies the geographic location of a device, object, or individual.
    - Geolocation data are stored in SQL, NoSQL and Databases specialized for geolocation data such as PostGIS (extension of PostgreSQL), Oracle Spatial, MongoDB with geospatial querying features

  - **Telemetry data**
    - Telematic data refers to information collected from remote devices and transmitted over telecommunications networks. This type of data is commonly associated with vehicles and includes a combination of telecommunication and monitoring technologies
  - **Images, Audio data**

  - **Automation data**

  - **…..**

More about timeseries data

- **Ordered by Time**
  - Each data point has a precise timestamp. The sequence matters — swapping two readings changes the meaning entirely.
- **Regular vs. Irregular Intervals**
  - Sensors may report every 10 s (regular) or only on-change (event-driven / irregular). Analysis methods differ accordingly.
- **Trend**
  - A long-term upward or downward movement in the data. Example: battery voltage gradually declining over weeks.
- **Seasonality**
  - Patterns that repeat at fixed intervals — daily, weekly, annual. Example: energy consumption peaks every weekday morning at 8:00.
- **High Volume & Velocity**
  - 1,000 sensors × every 10 s = 8.6 million data points per day. Storage and query performance must be purpose-built.
- **Autocorrelation**
  - A value at time t is often correlated with the value at t−1. Today's temperature predicts tomorrow's better than a random guess.
- **Missing & Late Data**
  - Network failures cause gaps. Packets can arrive out of order. Robust pipelines must handle imputation and late-arrival logic.
- **Append-Only Write Pattern**
  - New readings are always added at the end — historical values are rarely updated. Databases optimized for this (InfluxDB, TimescaleDB) outperform standard SQL for ingestion.

COMMON ANALYSIS FUNCTIONS - TIMESERIESDATA

FUNCTION	WHAT IT DOES	IOT EXAMPLE	PATH
Threshold / Rule Check	Compare current value against a fixed limit	Pressure > 8 bar → close valve	🔥 **Hot**
Tumbling Window Avg	Fixed-size non-overlapping time window, compute mean	Average power output per 15-min interval	🔥 **Hot** / 🌤️ **Warm**
Sliding Window	Overlapping window that moves forward with each new event	Detect if last 60 s avg vibration > threshold	🔥 **Hot**
Downsampling / Aggregation	Reduce resolution: many raw points → one summary value per interval	1-second readings → 5-min min/max/avg for dashboard	🌤️ **Warm**
Gap Filling / Interpolation	Fill missing timestamps with estimated values (linear, last-known)	Sensor offline for 3 min → fill gap before dashboard render	🌤️ **Warm**
Moving Average (SMA / EMA)	Smooth out short-term noise to reveal trend direction	7-day rolling avg of daily kWh consumption	🌤️ **Warm** / ❄️ **Cold**

COMMON ANALYSIS FUNCTIONS - TIMESERIESDATA

FUNCTION	WHAT IT DOES	IOT EXAMPLE	PATH
Anomaly Detection	Statistical or ML-based detection of outliers deviating from expected pattern	Motor current 3σ above baseline → bearing wear signal	🔥 **Hot** / ❄️ **Cold**
Seasonal Decomposition	Split a series into trend + seasonal component + residual noise	Separate daily usage pattern from long-term consumption growth	❄️ **Cold**
Forecasting (ARIMA / LSTM)	Predict future values based on historical patterns	Predict energy demand for next 30 days from 2 years of readings	❄️ **Cold**
Correlation Analysis	Measure how strongly two time series move together	Do open refrigeration doors correlate with compressor energy spikes?	❄️ **Cold**

WHERE TIME SERIES DATA IS CRITICAL — BY SECTOR

- **Manufacturing**
  - Vibration, temperature, and pressure on machines — monitored continuously to detect wear and prevent unplanned downtime (predictive maintenance)
- **Smart Grid / Energy**
  - Smart meters report consumption every 15 min. Load forecasting, demand-response programs, and grid balancing all depend on long time series histories.
- **Healthcare / Wearables**
  - ECG, blood oxygen, continuous glucose monitors produce dense time series. Arrhythmia detection and health trend analysis run on stream and on historical archives.
- **Precision Agriculture**
  - Soil moisture, pH, humidity sensors over growing seasons. Combined with weather APIs, ML models predict optimal irrigation timing and fertilizer application.
- **Transportation**
  - GPS tracks, axle load, and engine telemetry generate massive time series. Used for route optimization, predictive rail maintenance, and driver scoring.
- **Smart Buildings**
  - HVAC, lighting, occupancy, and air quality sensors logged continuously. Building management systems use time series to minimize energy waste while maintaining comfort.

Why Not Just Use a Standard Relational Database?

Purpose-built time series databases (InfluxDB, TimescaleDB, Azure Data Explorer and other timeseries dbs) are optimized for the IoT write pattern:

- millions of append-only inserts per day,
- range queries by time window,
- and automatic data retention policies
  - (e.g. keep raw data for 30 days, aggregates for 5 years).
- A standard SQL table with an unindexed timestamp column degrades dramatically at IoT scale.
- The Hot Path typically skips persistent storage altogether;
- The Warm Path uses a fast TSDB or columnar store;
- The Cold Path lands data in a compressed format (Parquet) in object storage for batch analysis.